

Shizheng Wen 闻仕政

PhD Student in AI & Applied Mathematics, ETH Zürich

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🌐 shizheng-wen.github.io

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Research Interests

Scientific computing and machine learning, with a focus on fast, scalable, and differentiable solvers for partial differential equations (PDEs): differentiable numerical methods, neural operator architectures for complex geometries, and large-scale PDE foundation models.

Education

ETH Zürich

2025 – Present

Ph.D. in Computational Mathematics and AI

Seminar for Applied Mathematics (SAM), Department of Mathematics; affiliated PhD, ETH AI Center.

Advisor: Prof. Siddhartha Mishra.

ETH Zürich

2022 – 2025

M.Sc. in Computational Science and Engineering 🔗 [course categories & personal notes](#)

Department of Mathematics. **Advisor:** Prof. Siddhartha Mishra.

Awarded the **ETH Medal** for an outstanding master's thesis.

Duke University

2019

Visiting Undergraduate

Aeroelasticity Group, Department of Mechanical Engineering and Materials Science.

Advisor: Prof. Earl Dowell.

Nanjing University of Aeronautics and Astronautics (NUAA)

2016 – 2020

B.Eng. in Energy and Power Engineering (with distinction)

School of Energy and Power Engineering. **Advisor:** Prof. Xianglei Liu.

Open-Source Project

- **TensorMesh** — creator. A fast, differentiable, JIT-free, debugging-friendly, GPU-native finite-element library for PyTorch.
- **torch-sla** — creator. A memory-efficient, differentiable sparse linear-algebra / linear-solver library for PyTorch with multiple backends.

Publications

1. **Preconditioned Physics-Informed Neural Operator Training.**
Shizheng Wen*, Siddhartha Mishra, Marius Zeinhofer*.
arXiv preprint, 2026. [\[arXiv\]](#) [\[code\]](#)
2. **Does Transolver really need a Transformer?**
Shizheng Wen, Siddhartha Mishra.
arXiv preprint, 2026. [\[arXiv\]](#) [\[code\]](#)
3. **Learning, Solving and Optimizing PDEs with TensorGalerkin: an Efficient High-Performance Galerkin Assembly Algorithm.**
Shizheng Wen*, Mingyuan Chi*, Tianwei Yu, Ben Moseley, Mike Yan Michelis, Pu Ren, Hao Sun, Siddhartha Mishra.
ICML, 2026. [\[arXiv\]](#) [\[code\]](#) [\[project\]](#)
4. **torch-sla: Differentiable Sparse Linear Algebra with Adjoint Solvers and Sparse Tensor Parallelism for PyTorch.**

Mingyuan Chi*, **Shizheng Wen***.

ICML Workshop (AI4Physics), 2026. [[arXiv](#)] [[code](#)] [[project](#)]

5. **MOSIV: Multi-Object System Identification from Videos.**

Chunjiang Liu, Xiaoyuan Wang, Qingran Lin, Albert Xiao, Haoyu Chen, **Shizheng Wen**, Hao Zhang, Lu Qi, Ming-Hsuan Yang, Laszlo A. Jeni, Min Xu, Yizhou Zhao.

ICLR, 2026. [[arXiv](#)] [[code](#)]

6. **Geometry Aware Operator Transformer as an Efficient and Accurate Neural Surrogate for PDEs on Arbitrary Domains.**

Shizheng Wen, Arsh Kumbhat, Levi Lingsch, Sepehr Mousavi, Yizhou Zhao, Praveen Chandrashekar, Siddhartha Mishra.

NeurIPS, 2025. [[arXiv](#)] [[code](#)] [[project](#)]

7. **RIGNO: A Graph-based Framework for Robust and Accurate Operator Learning for PDEs on Arbitrary Domains.**

Sepehr Mousavi, **Shizheng Wen**, Levi Lingsch, Maximilian Herde, Bogdan Raonić, Siddhartha Mishra.

NeurIPS, 2025. [[arXiv](#)] [[code](#)]

8. **Phase-field Simulation and Machine Learning of Low-field Magneto-elastocaloric Effect in a Multiferroic Composite.**

Wei Tang, **Shizheng Wen**, Huilong Hou, Qihua Gong, Min Yi, Wanlin Guo.

International Journal of Mechanical Sciences, 2024. [[doi](#)]

9. **Feature Identification in Complex Fluid Flows by Convolutional Neural Networks.**

Shizheng Wen, Michael W. Lee, Kai M. Kruger Bastos, Ian Eldridge-Allegra, Earl H. Dowell.

Theoretical and Applied Mechanics Letters, 2023. [[doi](#)]

10. **A Machine Learning Strategy for Modeling and Optimal Design of Near-field Radiative Heat Transfer.**

Shizheng Wen, Chunzhuo Dang, Xianglei Liu.

Applied Physics Letters, 2022. [[doi](#)]

11. **High-performance Three-body Near-field Thermophotovoltaic Energy Conversion.**

Chunzhuo Dang, Xianglei Liu, Haifeng Xia, **Shizheng Wen**, Qiao Xu.

Journal of Quantitative Spectroscopy and Radiative Transfer, 2021. [[doi](#)]

12. **Ultrahigh Thermal Rectification Based on Near-field Thermal Radiation between Dissimilar Nanoparticles.**

Shizheng Wen, Xianglei Liu, Sheng Cheng, Zhoubing Wang, Shenghao Zhang, Chunzhuo Dang.

Journal of Quantitative Spectroscopy and Radiative Transfer, 2019. [[doi](#)]

* Equal contribution.

Honors & Awards

• ICML Gold Reviewer	2026
• ETH Medal (top 2.5%)	2026
• National Scholarship (top 1%)	2019
• NUAA Presidential Fellowship (top 0.1%)	2019

Invited Talks

• <i>Differentiable Physics Simulation Engine: The infrastructure for physics foundation model.</i> Hong Kong University of Science and Technology (host: Tianju Xue).	Sep 2026
• <i>Accelerating computational science: a paradigm shift from sparse-iterative solver to dense-parallel computing.</i> Beihang University (host: Kun Wang).	Jun 2026
• <i>Learning, Solving and Optimizing PDEs with TensorGalerkin: an Efficient High-Performance Galerkin Assembly Algorithm.</i> Simon Fraser University (host: Wuyang Chen).	Mar 2026
• <i>Geometry Aware Operator Transformer as an Efficient and Accurate Neural Surrogate for PDEs on Arbitrary Domains.</i> Tsinghua University (host: Angelica Aviles-Rivero).	Mar 2026

- *Accelerating Computational Science: From Differentiable Solvers to Geometry Aware Operator Transformers*.
Stanford University (host: Charbel Farhat). Dec 2025

Teaching

- **Numerical Methods for Partial Differential Equations** (10 ECTS), 401-0674-00L.
ETH Zürich, Spring 2026. [[course page](#)]
- **AI in the Sciences and Engineering** (6 ECTS), 401-4656-21L.
ETH Zürich, Fall 2025 and Fall 2026. [[course page](#)]